VALIDATION REPORT

Inner Mongolia Wudaogou 50.25MW Wind Power Project in China

REPORT NO. 2007-0698

REVISION NO. 01

VALIDATION REPORT

DET NORSKE VERITAS
CERTIFICATIONAS

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Approved by: Michael Lehmann	Organisational unit: International Climate Change Services
Client: Chifeng Xinsheng Wind Power Co. Ltd.	Client ref.: Cui Baohe

Project Name: Inner Mongolia Wudaogou 50.25MW Wind Power Project
Country: China
Methodology: ACM0002
Version: 06
GHG reducing Measure/Technology: Wind Power
ER estimate: 142 848 **tCO_2e/year**
Size
☒ Large Scale
☐ Small Scale
Validation Phases:
☒ Desk Review
☒ Follow up interviews
☒ Resolution of outstanding issues
Validation Status
☐ Corrective Actions Requested
☐ Clarifications Requested
☒ Full Approval and submission for registration
☐ Rejected

In summary, it is DNV's opinion that the "Inner Mongolia Wudaogou 50.25MW wind power project" in China, as described in the PDD version 3.0 of 14 November 2007, meets all relevant UNFCCC requirements for the CDM and all relevant host country criteria and correctly applies the baseline and monitoring methodology ACM0002 version 06. DNV thus requests the registration of the project as a CDM project activity.

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Abbreviations

Explain any abbreviations that have been used in the report here.

BM	Build Margin
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CEF	Carbon Emission Factor
CER	Certified Emission Reduction
CL	Clarification request
CO_2	Carbon dioxide
CO_2e	Carbon dioxide equivalent
DNV	Det Norske Veritas
DNA	Designated National Authority
DRC	Development and Reform Commission
GHG	Greenhouse gas(es)
GWP	Global Warming Potential
IPCC	Intergovernmental Panel on Climate Change
LOA	Letter of Approval.
MP	Monitoring Plan
MVP	Monitoring and Verification Plan
NEPG	Northeast power grid
NGO	Non-governmental Organisation
ODA	Official Development Assistance
OM	Operating Margin
PDD	Project Design Document
UNFCCC	United Nations Framework Convention on Climate Change
SCE	Standard coal equivalent

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1 EXECUTIVE SUMMARY – VALIDATION OPINION

Det Norske Veritas Certification AS (DNV) has performed a validation of the "Inner Mongolia Wudaogou 50.25MW wind power project" in China. The validation was performed on the basis of UNFCCC criteria for the Clean Development Mechanism and host country criteria, as well as criteria given to provide for consistent project operations, monitoring and reporting.

The review of the project design documentation and the subsequent follow-up interviews have provided DNV with sufficient evidence to determine the fulfillment of stated criteria.

The host party is China and the Annex I party is the United Kingdom. Both countries fulfill the participation criteria and have approved the project and authorized the project participants. The DNA of China has confirmed that the project assist in achieving sustainable development.

The validation did not reveal any information that indicates that the project can be seen as a diversion of official development assistance (ODA) funding towards China.

The project correctly applies ACM0002 version 06: "Consolidated baseline & monitoring methodology for grid-connected electricity generation from renewable sources".

By generating renewable energy the project will displace fossil fuel based grid electricity. The project results in reductions of CO_2 emissions that are real, measurable and give long-term benefits to the mitigation of climate change. It is demonstrated that the project is not a likely baseline scenario. Emission reductions attributable to the project are hence additional to any that would occur in the absence of the project activity.

The total emission reductions from the project are estimated to be on the average 142 848 tCO2e per year over the first 7-year crediting period. The emission reduction forecast has been checked, and it is deemed likely that the stated amount is achieved given that the underlying assumptions do not change.

Adequate training and monitoring procedures have been implemented.

In summary, it is DNV's opinion that the "Inner Mongolia Wudaogou 50.25MW wind power project" in China as described in the PDD version 3.0 of 14 November 2007 meets all relevant UNFCCC requirements for the CDM and all relevant host country criteria and correctly applies the baseline and monitoring methodology ACM0002 version 06. DNV thus requests the registration of the "Inner Mongolia Wudaogou 50.25MW wind power project" as a CDM project activity.

2 INTRODUCTION

Chifeng Xinsheng Wind Power Co. Ltd. Has commissioned Det Norske Veritas Certification AS (DNV) to perform a validation of the “Inner Mongolia Wudaogou 50.25MW wind power project” in China (hereafter called “the project”). This report summarizes the findings of the validation of the project, performed on the basis of UNFCCC criteria for CDM projects, as well as criteria given to provide for consistent project operations, monitoring and reporting. UNFCCC criteria refer to Article 12 of the Kyoto Protocol, the CDM modalities and procedures and the subsequent decisions by the CDM Executive Board.

2.1 Objective

The purpose of a validation is to have an independent third party assess the project design. In particular, the project’s baseline, monitoring plan, and the project’s compliance with relevant UNFCCC and host Party criteria are validated in order to confirm that the project design, as documented, is sound and reasonable and meets the identified criteria. Validation is a requirement for all CDM projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of certified emission reductions (CERs).

2.2 Scope

The validation scope is defined as an independent and objective review of the project design document (PDD). The PDD is reviewed against the criteria stated in Article 12 of the Kyoto Protocol, the CDM modalities and procedures as agreed in the Marrakech Accords and the relevant decisions by the CDM Executive Board, including the approved baseline and monitoring methodology. The validation team has, based on the recommendations in the Validation and Verification Manual employed a risk-based approach, focusing on the identification of significant risks for project implementation and the generation of CERs.

The validation is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

3 METHODOLOGY

The validation consisted of the following three phases:

I a desk review of the project design documents

II follow-up interviews with project stakeholders

III the resolution of outstanding issues and the issuance of the final validation report and opinion

The following sections outline each step in more detail.

3.1 Desk Review of the Project Design Documentation

The following table outlines the documentation reviewed during the validation:

/1/ Dr. Zhang Xiaohua, Global Climate Change Institute of Tsinghua University and China Fulin Wind Power Development Corporation (Consultants), Project Design Document for the "Inner Mongolia Wudaogou 50.25MW Wind Power Project", Version 1.0, 10 July 2006 and version3.0 of 14 November 2007

/2/ Letter of Approval issued by DNA of China on 13 December 2006.

/3/ Letter of Approval issued by DNA of UK on 13 April 2007.

/4/ The feasibility study report of Inner Mongolia Wudaogou 50.25MW Wind Power Project in August 2005 and the approval letter by the Development and Reform Commission of Inner Mongolia Autonomous Region on 18 November 2005.

/5/ The EIA of the Inner Mongolia Wudaogou 50.25MW Wind Power Project on 25 August 2005 and the approval letter by the Environmental Protection Bureau of Inner Mongolia Autonomous Region on 20 September 2005.

/6/ Monitoring plan – Annex 4 in the PDD, version3.0 of 14 November 2007

/7/ Registration form of the stakeholder consultation meeting.

/8/ Permit for construction of the project, April 10, 2006

/9/ International Emission Trading Association (IETA) & the World Bank's Prototype Carbon Fund (PCF): *Validation and Verification Manual*. http://www.vvmanual.info

/10/ ACM0002 "Consolidated methodology for grid-connected electricity generation from renewable sources" version 06 of 19 May 2006.

/11/ CDM Executive Board: Tool for the demonstration and assessment of additionality, version 03

/12/ China Electric Power Yearbooks 1999, 2004 – 2006.

/13/ China Energy statistics Yearbooks 2004 – 2006.

/14/ Chinese National Standard GB2589-81

/15/ CDM EB, Answer to DNV's request for deviation of Chinese project activities from AM0005, received on 1 December 2005. To be found on

http://cdm.unfccc.int/Projects/Deviations

/16/ Revised 2006 IPCC Guidelines for National Greenhouse Gas Inventories

/17/ State Power Corporation of China. Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects. Beijing: China Electric Power Press, 2003

/18/ Investment increase statement by Chifeng Development and Reform Committee. 9 December 2005.

/19/ cdm.ccchina.gov.cn/WebSite/CDM/UpFile/File1364.pdf (Grid determination)

/20/ cdm.ccchina.gov.cn/WebSite/CDM/UpFile/File1365.pdf (BM calculation)

/21/ cdm.ccchina.gov.cn/WebSite/CDM/UpFile/File1358.xls (OM calculation)

/22/ Loan permission letter by China Development Bank, 9 March 2006.

/23/ Wind turbines purchase contract, 28 December 2005.

Main changes between the version published for the 30 days stakeholder commenting period and the final version submitted for registration:

- Changes related to the CAR and CL identified in the DNV's draft validation report.

- The version of PDD format has been changed from version 02 to version 03.

-Additionality tool has been changed from version 2 to 3.

3.2 Follow-up Interviews with Project Stakeholders

Personnel who have been interviewed and/or provided additional information to the presented documentation:

	Date	Name	Organization	Topic
/24/	2007-02-05	Mr. Cui Baohe, Deputy director of plan and operation department, Mr. Wang Kefang, Chief Financial Officer	Chifeng Xinsheng Wind Power Co. Ltd.	- Project background information. - Project technology, operation, maintenance and monitoring capability. - Project additionality. - Project monitoring and management plan. - Project approval status (incl. EIA approval, CDM project approval status) - Stakeholder consultation process.
/25/	2007-02-05	Ms. Feng Tianfeng,	China Fulin Wind Power	- Applicability of

Project manager	Development Corporation (consultant)	selected methodology. - Baseline determination. - Emission reductions calculation. - Emission reduction monitoring plan.

3.3 Resolution of Outstanding Issues

The objective of this phase of the validation was to resolve any outstanding issues which needed be clarified prior to DNV's positive conclusion on the project design. In order to ensure transparency a validation protocol was customised for the project. The protocol shows in transparent manner criteria (requirements), means of verification and the results from validating the identified criteria. The validation protocol serves the following purposes:

- It organises, details and clarifies the requirements a CDM project is expected to meet;
- It ensures a transparent validation process where the validator will document how a particular requirement has been validated and the result of the validation.

The validation protocol consists of two tables. The different columns in these tables are described in the figure below. The completed validation protocol for the Inner Mongolia Wudaogou 50.25MW Wind Power Project is enclosed in Appendix A to this report.

Findings established during the validation can either be seen as a non-fulfilment of CDM criteria or where a risk to the fulfilment of project objectives is identified. Corrective action requests (CAR) are issued, where:

i) mistakes have been made with a direct influence on project results;
ii) CDM and/or methodology specific requirements have not been met; or
iii) there is a risk that the project would not be accepted as a CDM project or that emission reductions will not be certified.

A request for clarification (CL) may be used where additional information is needed to fully clarify an issue.

Validation Protocol Table 1: Mandatory Requirements for CDM Project Activities		
Requirement	***Reference***	***Conclusion***
The requirements the project must meet.	*Gives reference to the legislation or agreement where the requirement is found.*	*This is either acceptable based on evidence provided (**OK**), a **Corrective Action Request (CAR)** of risk or non-compliance with stated requirements or a request for **Clarification (CL)** where further clarifications are needed.*

Validation Protocol Table 2: Requirement checklist				
Checklist Question	***Reference***	***Means of verification (MoV)***	***Comment***	***Draft and/or Final Conclusion***
The various requirements in Table 2 are linked to checklist questions the project should meet. The checklist is organised in different sections, following the logic of the large-scale PDD template, version 03 - in effect as of: 28 July 2006. Each section is then further sub-divided.	*Gives reference to documents where the answer to the checklist question or item is found.*	*Explains how conformance with the checklist question is investigated. Examples of means of verification are document review (DR) or interview (I). N/A means not applicable.*	*The section is used to elaborate and discuss the checklist question and/or the conformance to the question. It is further used to explain the conclusions reached.*	*This is either acceptable based on evidence provided (**OK**), or a **corrective action request (CAR)** due to non-compliance with the checklist question (See below). A request for clarification (CL) is used when the validation team has identified a need for further clarification.*

Validation Protocol Table 3: Resolution of Corrective Action and Clarification Requests			
Draft report clarifications and corrective action requests	***Ref. to checklist question in table 2***	***Summary of project owner response***	***Validation conclusion***
If the conclusions from the draft Validation are either a CAR or a CL, these should be listed in this section.	*Reference to the checklist question number in Table 2 where the CAR or CL is explained.*	*The responses given by the project participants during the communications with the validation team should be summarised in this section.*	*This section should summarise the validation team's responses and final conclusions. The conclusions should also be included in Table 2, under "Final Conclusion".*

Figure 1 Validation protocol tables

3.4 Internal Quality Control

The draft validation report including the initial validation findings underwent a technical review before being submitted to the project participants. The final validation report underwent another technical review before requesting registration of the project activity. The technical review was performed by a technical reviewer qualified in accordance with DNV's qualification scheme for CDM validation and verification.

3.5 Validation Team

Role/Qualification	Last Name	First Name	Country
Team leader GHG auditor	Yang	Weidong	China
CDM validator	Yue	Mindy	China
GHG auditor	Kuo	Tim	China
Technical reviewer (applicant)	Flagstad	Ole Andreas	Norway
Technical reviewer	Rescalvo	Miguel	Norway
Sector expert	Lehmann	Michael	Norway

The qualification of each individual validation team member is detailed in Appendix B to this report.

4 VALIDATION FINDINGS

The findings of the validation are stated in the following sections. The validation criteria (requirements), the means of verification and the results from validating the identified criteria are documented in more detail in the validation protocol in Appendix A.
The final validation findings relate to the project design as documented and described in the revised and resubmitted project design documentation.

4.1 Participation Requirements

The project participants are Chifeng Xinsheng Wind Power Co. Ltd. of China and EDF Trading Limited of the United Kingdom. The host Party China and the participating Annex I Party the United Kingdom meet the requirements to participate in the CDM.

The DNA of China has issued a Letter of Approval (LoA) /2/ on 13 December 2006, authorizing Chifeng Xinsheng Wind Power Co. Ltd as project participant and also confirmed that the project assists in achieving sustainable development.

The DNA of the United Kingdom has issued a LoA /3/ on 13 April 2007, authorizing EDF Trading Limited as project participant.

The validation did not reveal any information that indicates that the project can be seen as a diversion of official development assistance (ODA) funding towards China.

4.2 Project Design

The project involves installation and operation of 67 wind turbines in Wengniute County, Inner Mongolia Autonomous Region, China.

The installed capacity of each unit is 750 kW, thus constituting a total generation capacity of 50.25MW. The installation also includes a central control room for control, measurement and surveillance of the wind farm **/1/**. The project proposes the installation of Xinjiang Gold Wind S48/750 wind turbines, which are domestically made and used to some scale in China, representing state of the art technology. The technology is deemed to reflect current good practice.

Being a renewable electricity project, the project activity will generate greenhouse gas (GHG) emission reductions by avoiding CO_2 emissions from electricity generation by fossil fuel power plants.

The project's system boundaries are clearly defined as the north-east power grid (NEPG).

The project activity started on 10 April 2006, date when the construction permit was obtained. /8/. The expected operational lifetime of the project activity is 20 years. A renewable crediting period of 7 years has been chosen for the project, starting from 1 March 2008 The emission reductions are estimated to be 142 848 tCO_2e per year and 999 936 tCO_2e over the first seven-year crediting period.

4.3 Baseline Determination

The project applies the approved baseline methodology ACM0002 (version 06), titled "Consolidated methodology for grid-connected electricity generation from renewable sources".

The applied baseline methodology is justified as it has been demonstrated that the project activity ensures that:

- It is a grid connected zero emission renewable power generation activity from wind energy.
- The project does not involve switching from fossil fuel to renewable energy at the project site.

The project boundary is defined as the site of the project activity and all power plants connected physically to the northeast power grid including the Heilongjiang, Jilin and Liaoning provincial grids to which the project is connected. This is in line with the delineation of grid boundaries as provided by the DNA of China /19/. The defined project boundary is in line with ACM0002 (version 06).

Emission sources and gases included in the project boundary are:

	GHGs involved	*Description*
Baseline emissions	*CO_2*	*The Northeast Power Grid (China)*
Project emissions	*N/A*	*Project emission is regarded as zero as the project is a renewable energy (wind source) project.*
Leakage	*N/A*	*There are no leakages that need to be considered in applying this methodology.*

In the baseline scenario the electricity delivered from the project activity to the grid would have been generated by fossil fuels grid-connected power plants and by the addition of new generation sources. This is reflected in the combined margin (CM) - the weighted average of the operating Margin (OM) emission factor and the build margin (BM) emission factor. The weighting is set to respectively 75% and 25%, the default values stipulated by ACM0002 version 06 for wind farm projects.

The NEPG is dominated by coal-fired power plants. It is deemed likely that coal-fired power plants will continue to dominate the power sector due to the local availability of low-cost coal. It is expected that renewable capacity additions will not have significant effects on the mix of the NEPG during the first crediting period.

The baseline determination is transparent and reasonable.

4.4 Additionality

The additionality of the project has been established using the "*Tool for the demonstration and assessment of additionality*" version 03 approved by the CDM-EB in the EB 29th meeting.

The project activity started on 10 April 2006, date when the construction permit was obtained. /8/. It is DNV's opinion that this date correctly represent the earliest of the dates at which the implementation or construction or real action of the project activity begun.

DNV has verified that the project owner considered the CDM benefits prior to the starting date of the project, as this is confirmed in the loan permission letter of China Development Bank on 9 March 2006 /22/. In addition, the approval letter of the feasibility study report of the project by the Development and Reform Commission of Inner Mongolia Autonomous Region on 18 November 2005 /4/ refers to CDM.

Step 1: Identification of the alternatives to the project activity consistent with the current laws and regulations.

The alternate baseline scenarios for the project activity have been suitably identified as,

a) The thermal power plant with the same capacity or the same annual electricity output as the proposed project;
b) The proposed project not undertaken as a CDM project activity but as a commercial project;
c) The other renewable energy power plant with the same capacity or the same annual electricity output as the proposed project;
d) The northeast power grid as the provider for the same capacity and electricity output as the proposed project.

The fossil fuel-fired power plant scenario does not comply with the Chinese law as thermal power plants with a capacity less than 135 MW are prohibited to be built in areas covered by large grids such as provincial grids[*]. The project owner, Chifeng Xinsheng Wind Power Co., Ltd. is dedicated to wind farm business. Other renewable energy power plant such as hydropower hydro power plant may not be alternatives for an independent power producer investing in wind energy. The proposed project activity not undertaken as a CDM project activity is not a realistic and credible alternative, as discussed in the analysis below. The only realistic and credible alternative for the proposed project is thus a comparable capacity or electricity generation addition provided by NEPG.

Step 2: Investment analysis.

As the proposed project generates financial and economic benefits through the sales of electricity other than CDM related income, a simple cost analysis (option I of the methodology) can not be applied. The alternative for the baseline scenario of the proposed project is not a similar investment project, so option II is also not an appropriate choice. Hence, a benchmark analysis (option III of Step 2 of tool for the demonstration and assessment of additionality) is selected for conducting the investment analysis.

In China an IRR of 8 % for the total investment of a project is regarded as a benchmark[†] for investments in hydropower plants, fossil fuel fired plants and wind farm projects. Based on the data in the feasibility study report /4/ and the investment increase statement approved by Chifeng Development and Reform Committee /18/, the project IRR without CER revenues is 7.07 %, which shows that the project is not financially attractive compared to the benchmark in the absence of CDM benefits.

A sensitivity analysis shows the changes in the total investment or power tariff can have a significant impact on project financial performance:

[*] Notice on Strictly Prohibiting the Installation of Fuel-fired Generators with the Capacity of 135MW or below issued by the General Office of the State Council, decree No. 2002-6.

[†] State Power Corporation of China. Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects. Beijing: China Electric Power Press, 2003

- If the total investment decreases by 6.8%, the IRR of the project could exceed the benchmark. However it is not likely that the total investment will decrease by 6.8% for the majority of the investment is related to the purchase of wind turbines, whose price is rather stable. Furthermore, the project investment has already been committed, the wind turbines purchase contract has been verified /23/ and the investment was as the investment increase statement.
- If the electricity tariff increases by about 7.2 %, the IRR of the project could exceed also the benchmark. However there is extremely unlikely for the tariff of the proposed project to have an increase of 7.2%. According to China's Management Rules on Tariff issued by NDRC‡, the tariff of the un-tendering projects should be determined by the government with reference to the tariff of tendering wind projects. By this pricing principle, China government is gradually lowering down the wind power in-grid tariff§. The highest tariff of the approved tendering wind power project is 0.5190 Yuan/kWh (incl. VAT) before 2006**, which is lower than the expected tariff in the Feasibility Study Report of the proposed project, so the final approved tariff for the proposed project should be lower than the tariff in the Feasibility Study Report of the proposed project. And, once the tariff is issued, it will be strictly regulated by the government, neither the project owner nor the grid company can change it.
- If the plant load factor (PLF) increase by about 7.2%, the IRR of the project could also exceed the benchmark. However, considering the PLF value is positive correlation with the wind speed, and the PLF value in the feasibility study report is based on the wind speed data of 2005. The data provided by local meteorological station shows that the average wind speed in 2005 is 2.9m/s, which is higher than the average wind speed 2.8m/s of the latest 30 years. The wind speed exceeds 3.1m/s when the wind speed is 7.2% higher than 2.9m/s. However, the average wind speed of the project site tends to decrease and it is 2.8 m/s over the past 30 years (from 1975 to 2004) /4/. The probability the speed beyond 3.1m/s is very small in the latest 10 years. Therefore, it is very hard that the PLF is 7.2% higher than the 28.7% assumption to the proposed project.
- The IRR for total investment will remain below the benchmark while the O&M cost varies in the range of -10%-+10%.

In conclusion, the investment analysis and sensitivity assessment have shown that the project activity is unlikely to be the most financially attractive option.

Step 3: Barrier analysis.

Barrier analysis has not been selected to demonstrate the additionality.

Step 4: Common practice analysis.

The common practice analysis is done in Inner Mongolia Autonomous Region. Projects developed within the same region face a similar regulatory framework that makes them comparable. The common practice shows that earlier wind farm in Inner Mongolia Autonomous Region enjoyed a very favorable electricity tariff (RMB 0.76Yuan/kWh) that is impossible for nowadays wind farm projects. Other similar wind farms have all applied for

‡ Interim Regulation for Tariff of Renewable Energy Power Generation and Appointment of Expenses FAGAIJIAGE(2006) No.7

§ http://www.eri.org.cn/manage/upload/uploadimages/eri200672795944.pdf

** http://news.cepee.com/text.php?NewsId=25998

CDM projects due to the same financial unattractiveness and investment barrier as the proposed project activity. Four of these projects have been successfully registered as CDM projects. The references of the listed similar projects and the reference of the wind farm which enjoyed favorable electricity tariff have been verified and are deemed to be reliable. Therefore the project activity cannot be said to represent common practice.

In summary, it is sufficiently demonstrated that the project is not a likely baseline scenario and that emission reductions occurring from this will hence be additional.

4.5 Monitoring

The project applies the approved monitoring methodology ACM0002 version 06 "Consolidated monitoring methodology for zero emissions grid-connected electricity generation from renewable sources". The selected monitoring methodology is applicable for the project activity as it involves grid-connected renewable power generation using wind energy.

4.5.1 Parameters determined ex-ante

The combined margin emission factor is determined *ex-ante* based on the most recent information available. More detailed information is provided below

4.5.2 Parameters monitored ex-post

The methodology requires monitoring of the following for wind farm projects:

• Electricity generation from the proposed project activity;

• Data needed to recalculate the operating margin emission factor, if needed, based on the choice of the method to determine the operating margin (OM), consistent with "Consolidated baseline methodology for grid-connected electricity generation from renewable sources" (ACM0002);

• Data needed to recalculate the build margin emission factor, if needed, consistent with "Consolidated baseline methodology for grid-connected electricity generation from renewable sources" (ACM0002);

The operating margin and build margin emission factor are determined *ex-ante*, therefore the parameter monitored *ex-post* is the electricity generation from the proposed project activity. The net electricity generated from the project will be measured continuously and recorded monthly. This data will be cross verified against the sales receipt from the grid.

4.5.3 Management system and quality assurance

The project's monitoring plan **/1/** includes:

- A description of the responsibilities and authorities for project management,
- Procedures for monitoring and reporting, and QA/QC procedures,
- A description of the installation of metering equipment,
- Procedures for the calibration of metering equipment,
- A description of training and maintenance needs.

Detailed procedures have been elaborated and are in place. These will be maintained and implemented to enable subsequent verification of emission reductions.

4.6 Estimate of GHG Emissions

The emission reduction ERy by the project activity during the crediting period is the difference between baseline emissions (BEy), project emissions (PEy) and emissions due to leakage (Ly), as follows:

1) Baseline emissions: baseline emissions (BEy in tCO_2) are the product of the baseline emissions factor (EFy in tCO_2/MWh) for the China northeast power grid (NEPG) times the electricity supplied by the project activity to the grid (EGy in MWh)

2) Project emissions: there are no emissions from the project which is a renewable energy project

3) Leakage: no leakage has to be considered for the proposed project activity

4) Emission reduction: ERy= BEy- PEy- Ly= BEy

The baseline emission factor for the project is determined *ex-ante* as a combined margin, consisting of combination of the operating margin (OM) and build margin (BM).

For the calculation of the OM emission factor, the simple OM emission factor calculation method is selected because low cost must run projects constitute less than 50% of the total grid generation and data is not available for applying the dispatch data analysis.

The aggregated generation and fuel consumption data are used due to less aggregated data are not available in the NEPG. Country specific data for net calorific value (NCV_i) of each type of fossil fuel, the IPCC 2006 default values for the oxidation factor of each type of fossil fuel and the total electricity delivered to the NEPG are selected and are deemed reasonable. Vintage data for the years 2003, 2004 and 2005 from China Energy Statistics Yearbooks and China Electric Power Yearbooks 2004-2006 /13/ /12/ editions are used for the operating margin calculation. The OM is correctly evaluated to be 1.2402 tCO_2/MWh.

For the calculation of the build margin (BM) emission factor, because plant specific fuel consumption and electricity generation data is not public available in China, DNV requested guidance from the CDM Executive Board for a deviation of the baseline methodology of AM0005 and received the following answers (Since AM0005 was replaced by ACM0002, the deviation confirmed by EB is deemed to be applicable for this project).

- Use of capacity additions for estimating the build margin emission factor for grid electricity.
- Use of weights estimated using installed capacity in place of annual electricity generation.
- Use the efficiency level of the best technology commercially available in the provincial/regional or national grid of China, as a conservative proxy, for each fuel type in estimating the fuel consumption to estimate the build margin (BM).

Following the EB's guidance the build margin is calculated as follows:

- The capacity additions from the years 1998 to 2005 /12/ is chosen and reached 21.34% of total installed capacity.
- The weight of installed capacity additions for thermal power plant is accounted for 91.31% of total installed capacity additions;
- There are no data available of installed capacity additions for oil and gas power in NEPG. However China Energy Statistics Yearbook (2006) /13/ shows that the oil and gas used in NEPG are very small, and only for starting up systems of coal fired power plant,

accounting for ca. 3.29% of the total CO_2 emissions. The installed capacity addition for coal, oil and gas power plants being regarded as proportional with their CO_2 emissions percentage is deemed reasonable.

- The coal consumption efficiency of 343.33 gSCE/kWh is selected as the best technology commercially available in China. It can be acknowledged as the best available data available for estimating the BM in the China /20/. This best technology corresponds to a 35.82% of power supply efficiency for coal-fired electricity generation. The gas and oil consumption efficiency of 258 gSCE/kWh is selected as the best technology commercially available in China /20/ It can be acknowledged as the best available data available for estimating the BM in the China. This best technology corresponds to a 47.67% of power supply efficiency for gas or oil-fired electricity generation.
- The IPCC 2006 value of 25.8 tC/TJ and a carbon oxidation factor of 100% are used to calculate the BM emission coefficients.
- The BM is calculated to be 0.8632 tCO_2/MWh

The weights ω_{OM} and ω_{BM} are selected as 0.75 and 0.25, respectively, as stipulated for wind project by ACM0002 (version 06). The combined margin of 1.1459 tCO_2/MWh is fixed *ex-ante* for the entire first crediting period.

The GHG calculations are complete and transparent, and their accuracy has been verified.

4.7 Environmental Impacts

An Environmental Impact Assessment (EIA) has been conducted according to Chinese law & regulation. The potential environmental impacts have been sufficiently identified. No significant environmental impacts are expected from the project activity. The Environmental Protection Bureau of Inner Mongolia Autonomous Region approved the project activity on 29 September 2006 /5/.

4.8 Comments by Local Stakeholders

Besides the stakeholder consultation process stipulated in the Chinese EIA regulation, the project developer has conducted an additional stakeholder consultation. On 26 July 2006, the project owner held a stakeholder meeting in Wengniute County /7/. Stakeholder representatives from the local Development and Reform Bureau, the local Environmental Protection Bureau, local Power Supply Corporation and Yangshugoumen Village in Wengniute County attended the meeting. There were no adverse comments on the project activity. A summary of comments is provided and has been verified by DNV.

4.9 Comments by Parties, Stakeholders and NGOs

The PDD of 10 July 2006 was made publicly available on DNV's climate change website (www.dnv.com/certification/climatechange) and Parties, stakeholders and NGOs were through the CDM website invited to provide comments during a 30 days period from 26 August 2006 to 24 September 2006. No comment was received during this period.

APPENDIX A

CDM VALIDATION PROTOCOL

Table 1 Mandatory Requirements for Clean Development Mechanism (CDM) Project Activities

Requirement	Reference	Conclusion
About Parties		
The project shall assist Parties included in Annex I in achieving compliance with part of their emission reduction commitment under Art. 3.	Kyoto Protocol Art.12.2	OK
The project shall assist non-Annex I Parties in contributing to the ultimate objective of the UNFCCC.	Kyoto Protocol Art.12.2.	OK
The project shall have the written approval of voluntary participation from the designated national authority of each Party involved.	Kyoto Protocol Art. 12.5a, CDM Modalities and Procedures §40a	~~CAR 1~~ OK
The project shall assist non-Annex I Parties in achieving sustainable development and shall have obtained confirmation by the host country thereof.	Kyoto Protocol Art. 12.2, CDM Modalities and Procedures §40a	~~CAR 1~~ OK
In case public funding from Parties included in Annex I is used for the project activity, these Parties shall provide an affirmation that such funding does not result in a diversion of official development assistance and is separate from and is not counted towards the financial obligations of these Parties.	Decision 17/CP.7, CDM Modalities and Procedures Appendix B, § 2	OK
Parties participating in the CDM shall designate a national authority for the CDM.	CDM Modalities and Procedures §29	OK
The host Party and the participating Annex I Party shall be a Party to the Kyoto Protocol.	CDM Modalities §30/31a	OK
The participating Annex I Party's assigned amount shall have been calculated and recorded.	CDM Modalities and Procedures §31b	OK
The participating Annex I Party shall have in place a national system for estimating GHG emissions and a national registry in accordance with Kyoto Protocol Article 5	CDM Modalities and Procedures §31b	OK

Requirement	Reference	Conclusion
and 7.		
About additionality		
Reduction in GHG emissions shall be additional to any that would occur in the absence of the project activity, i.e. a CDM project activity is additional if anthropogenic emissions of greenhouse gases by sources are reduced below those that would have occurred in the absence of the registered CDM project activity.	Kyoto Protocol Art. 12.5c, CDM Modalities and Procedures §43	OK
About forecast emission reductions and environmental impacts		
The emission reductions shall be real, measurable and give long-term benefits related to the mitigation of climate change.	Kyoto Protocol Art. 12.5b	OK
For large-scale projects only		
Documentation on the analysis of the environmental impacts of the project activity, including transboundary impacts, shall be submitted, and, if those impacts are considered significant by the project participants or the Host Party, an environmental impact assessment in accordance with procedures as required by the Host Party shall be carried out.	CDM Modalities and Procedures §37c	OK
About stakeholder involvement		
Comments by local stakeholders shall be invited, a summary of these provided and how due account was taken of any comments received.	CDM Modalities and Procedures §37b	OK
Parties, stakeholders and UNFCCC accredited NGOs shall have been invited to comment on the validation requirements for minimum 30 days, and the project design document and comments have been made publicly available.	CDM Modalities and Procedures §40	OK
Other		
The baseline and monitoring methodology shall be previously approved by the CDM Executive Board.	CDM Modalities and Procedures §37e	OK

Requirement	Reference	Conclusion
A baseline shall be established on a project-specific basis, in a transparent manner and taking into account relevant national and/or sectoral policies and circumstances.	CDM Modalities and Procedures §45c,d	OK
The baseline methodology shall exclude to earn CERs for decreases in activity levels outside the project activity or due to force majeure.	CDM Modalities and Procedures §47	OK
The project design document shall be in conformance with the UNFCCC CDM-PDD format.	CDM Modalities and Procedures Appendix B, EB Decision	OK
Provisions for monitoring, verification and reporting shall be in accordance with the modalities described in the Marrakech Accords and relevant decisions of the COP/MOP.	CDM Modalities and Procedures §37f	OK

Table 2 Requirements Checklist

CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	**Ref.**	**MoV***	**COMMENTS**	**Draft Concl.**	**Final Concl.**
A. General Description of Project Activity *The project design is assessed.*					
Project Boundaries *Project Boundaries are the limits and borders defining the GHG emission reduction project.*					
Are the project's spatial boundaries (geographical) clearly defined?	**/1/**	DR I	Yes. The project is located in Wengniute County, Inner Mongolia Autonomous Region, China. The geographical coordinates are east longitude 117°54′-118°01′ and north latitude 42°39′-42°42′.		OK
Are the project's system boundaries (components and facilities used to mitigate GHGs) clearly defined?	**/1/**	DR	The northeast (regional) power grid (NEPG) is defined as the project system boundaries.		OK
Participation Requirements *Referring to Part A, Annex 1 and 2 of the PDD as well as the CDM glossary with respect to the terms Party, Letter of Approval, Authorization and Project Participant.*					
Which Parties and project participants are participating in the project?	**/1/**	DR	The project participants are Chifeng Xinsheng Wind Power Co. Ltd., China and EDF Trading Limited, the United Kingdom of Great Britain and Northern Ireland.		OK
Have all involved Parties provided a valid and complete letter of approval and have all private/public project participants been		DR	The letters of approval from the DNA of China and the UK have not been obtained.	~~CAR 1~~	OK

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CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	Ref.	MoV*	COMMENTS	Draft Concl.	Final Concl.
authorized by an involved Party?					
Do all participating Parties fulfil the participation requirements as follows: - Ratification of the Kyoto Protocol - Voluntary participation - Designated a National Authority		DR	China ratified the Kyoto Protocol on 30 August, 2002. The UK ratified the Kyoto Protocol on 31 May 2002. Both Parties participates in the CDM on a voluntary basis. Both Parties involved have designated national authorities for the CDM.		OK
Potential public funding for the project from Parties in Annex I shall not be a diversion of official development assistance.	**/1/**	DR	The validation did not reveal any information that indicates that the project can be seen as a diversion of official development assistance (ODA) funding towards the China.		OK
Technology to be employed *Validation of project technology focuses on the project engineering, choice of technology and competence/ maintenance needs. The validator should ensure that environmentally safe and sound technology and know-how is used.*					
Does the project design engineering reflect current good practices?	**/1/**	DR	Yes. The project design as developed in the Feseability Study Report engineering reflects current good practices in China.		OK
Does the project use state of the art technology or would the technology result in a significantly better performance than any commonly used technologies in the host country?	**/1/**	DR	Yes. The wind turbines applied are Goldwind S48-750kW wind turbines, which are manufactured domestically and have been		OK

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CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	Ref.	MoV*	COMMENTS	Draft Concl.	Final Concl.
			used in some scale. It uses state of the art technology.		
Does the project make provisions for meeting training and maintenance needs?	**/1/**	DR I	Yes. The training has been provided by project owner and manufacturer.		OK
Contribution to Sustainable Development *The project's contribution to sustainable development is assessed.*					
Has the host country confirmed that the project assists it in achieving sustainable development?		DR	The letter of approval from the DNA of China confirming the project being in line with the sustainable development policies of host country has not been received.	~~CAR 1~~	OK
Will the project create other environmental or social benefits than GHG emission reductions?	**/1/**	DR	Yes. The project will, among others benefits, mitigate local environmental pollution caused by coal-fired power plants and create local employment opportunity.		OK
B. Project Baseline *The validation of the project baseline establishes whether the selected baseline methodology is appropriate and whether the selected baseline represents a likely baseline scenario.*					
Baseline Methodology *It is assessed whether the project applies an appropriate baseline methodology.*					
Does the project apply an approved methodology and the correct version thereof?	**/1/** /10/	DR	Yes. The project applies ACM0002 (version 06) "Consolidated baseline methodology for grid-connected electricity generation from		OK

CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	Ref.	MoV*	COMMENTS	Draft Concl.	Final Concl.
			renewable sources".		
Are the applicability criteria in the baseline methodology all fulfilled?	**/1/** /10/	DR	Yes. The project is a capacity addition from a renewable energy source and does not involve on-site fuel switch from fossil fuels to a renewable source. The geographic and system boundaries for the relevant electricity grid (northeast power grid) can be clearly identified.		OK
Baseline Scenario Determination *The choice of the baseline scenario will be validated with focus on whether the baseline is a likely scenario, and whether the methodology to define the baseline scenario has been followed in a complete and transparent manner.*					
What is the baseline scenario?	**/1/**	DR	The baseline is determined as continued operation of the existing power plants and the addition of new generation sources to meet electricity demand.		OK
What other alternative scenarios have been considered and why is the selected scenario the most likely one?	**/1/**	DR	The alternates to the project activity have been suitably identified as, a) Construction of a thermal power plant with the same capacity or the same annual electricity output as the proposed project; b) The proposed project not undertaken as a CDM project activity but as a commercial project;		OK

CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	**Ref.**	**MoV***	**COMMENTS**	**Draft Concl.**	**Final Concl.**
			c) The other renewable energy power plant with the same capacity or the same annual electricity output as the proposed project; d) The North China Power Grid as the provider for the same capacity and electricity output as the proposed project. The baseline scenario has been identified according to ACM0002 version 06: The project activity does not modify or retrofit an existing electricity generation facility, therefore the baseline scenario is electricity delivered to the grid by the project would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.		
Has the baseline scenario been determined according to the methodology?	**/1/** /10/	DR	Yes.		OK
Has the baseline scenario been determined using conservative assumptions where possible?	**/1/**	DR	Yes.		OK
Does the baseline scenario sufficiently take into account relevant national and/or sectoral policies, macro-economic trends and	**/1/**	DR	Yes. The renewable energy law, sectoral policy and development trends in NEPG have		OK

CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	**Ref.**	**MoV***	**COMMENTS**	**Draft Concl.**	**Final Concl.**
political aspirations?			been taken into account.		
Is the baseline scenario determination compatible with the available data and are all literature and sources clearly referenced?	**/1/**	DR	Yes.		OK
Have the major risks to the baseline been identified?	**/1/**	DR	Yes. The major risk to the baseline is dramatic increase of power generation from renewable sources in future, such as wind and hydro.		OK
Additionality Determination *The assessment of additionality will be validated with focus on whether the project itself is not a likely baseline scenario.*					
Is the project additionality assessed according to the methodology?	**/1/** /11/	DR I	The project additionality is demonstrated by applying the "Tool for the demonstration and assessment of additionality" version 3. *Step 1: Identification of the alternatives to the project activity consistent with the current laws and regulations.* The alternates to the project activity have been suitably identified as, e) Construction of a thermal power plant with the same capacity or the same annual electricity output as the proposed project;		OK

CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	**Ref.**	**MoV***	**COMMENTS**	**Draft Concl.**	**Final Concl.**
			f) The proposed project not undertaken as a CDM project activity but as a commercial project; g) The other renewable energy power plant with the same capacity or the same annual electricity output as the proposed project; h) The North China Power Grid as the provider for the same capacity and electricity output as the proposed project. The only realistic and credible alternative for the proposed project has been demonstrated to be a comparable capacity or electricity generation addition provided by NEPG. *.Step 2: Investment analysis.* Benchmark analysis (Option III) is justified to conduct the investment analysis. - In China an IRR of 8 % for total investment of a project is regarded as benchmark for investments in hydropower plants, fossil fuel fired plants and wind farm projects. The project IRR without CER revenues is 7.07 % which shows that the project is not financially attractive in absence of CDM benefits. A sensitivity analysis has been carried out. It		

CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	**Ref.**	**MoV***	**COMMENTS**	**Draft Concl.**	**Final Concl.**
			has been demonstrated that it is unlikely that the critical parameters' values reach a level where the project can be considered financially attractive. In conclusion, the investment analysis and sensitivity assessment have shown that the project activity is unlikely to be the most financially attractive option. *Step 3: Barrier analysis.* Barrier analysis has not been selected to demonstrate additionality. *Step 4: Common practice analysis.* The common practice analysis is done in Inner Mongolia Autonomous Region. Projects developed within the same region face a similar regulatory framework that makes them comparable. The common practice shows that earlier wind farm in Inner Mongolia Autonomous Region enjoyed a very favourable electricity tariff (RMB 0.76Yuan/kWh) that is impossible for nowadays wind farm projects. Other similar wind farms have all applied for CDM projects due to the same financial unattractiveness and investment barrier as the proposed project activity. Four of these projects have been successfully registered as		

CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	**Ref.**	**MoV***	**COMMENTS**	**Draft Concl.**	**Final Concl.**
			CDM projects. The references of the listed similar projects and the reference of the wind farm which enjoyed favourable electricity tariff have been verified and are deemed to be reliable. Therefore the project activity cannot be said to represent common practice.		
Are all assumptions stated in a transparent and conservative manner?	**/1/**	DR	Yes.		OK
Is sufficient evidence provided to support the relevance of the arguments made?	**/1/**	DR	Yes.		OK
If the starting date of the project activity is before the date of validation, has sufficient evidence been provided that the incentive from the CDM was seriously considered in the decision to proceed with the project activity?	**/1/** /4/	DR	Evidence that the incentive from the CDM was seriously considered in the decision to proceed with the project activity needs to be provided.	~~CL 1~~	OK
Calculation of GHG Emission Reductions – Project emissions *It is assessed whether the project emissions are stated according to the methodology and whether the argumentation for the choice of default factors and values – where applicable – is justified.*					
Are the calculations documented according to the approved methodology and in a complete and transparent manner?	**/1/**	DR	Project emission is regarded as zero as the		OK

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CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	**Ref.**	**MoV***	**COMMENTS**	**Draft Concl.**	**Final Concl.**
	/10/		project is a renewable energy (wind source) project.		
Calculation of GHG Emission Reductions – Baseline emissions *It is assessed whether the baseline emissions are stated according to the methodology and whether the argumentation for the choice of default factors and values – where applicable – is justified.*					
Are the calculations documented according to the approved methodology and in a complete and transparent manner?	**/1/** /10/	DR	Yes. Baseline emissions are calculated as the net electricity generated form the renewable source times the NEPG emission factor. Auxiliary energy use for the operation of the plant is accounted for as the net electricity generated from the project is used for the calculations. The grid emission factor is correctly calculated in line with ACM0002 as a combined margin (CM), consisting of the combination of operating margin (OM) and build margin (BM).		OK
Have conservative assumptions been used when calculating the baseline emissions?	**/1/** /10/	DR	Yes.		OK
Are uncertainties in the baseline emission estimates properly addressed?	**/1/** /10/	DR	Yes.		OK

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CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	**Ref.**	**MoV***	**COMMENTS**	**Draft Concl.**	**Final Concl.**
Calculation of GHG Emission Reductions – Leakage *It is assessed whether leakage emissions are stated according to the methodology and whether the argumentation for the choice of default factors and values – where applicable – is justified.*					
Are the leakage calculations documented according to the approved methodology and in a complete and transparent manner?	**/1/** /10/	DR	There are no leakages that need to be considered in applying this methodology.		OK
Emission Reductions *The emission reductions shall be real, measurable and give long-term benefits related to the mitigation of climate change.*					
Are the emission reductions real, measurable and give long-term benefits related to the mitigation of climate change.	**/1/**	DR	Yes.		OK
Monitoring Methodology *It is assessed whether the project applies an appropriate baseline methodology.*					
Is the monitoring plan documented according to the approved methodology and in a complete and transparent manner?	**/1/** /10/	DR	The monitoring plan is in accordance with the approved monitoring methodology ACM0002 (version 06) "Consolidated monitoring methodology for grid-connected electricity generation from renewable sources" and is in a complete and transparent manner.		OK

CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	**Ref.**	**MoV***	**COMMENTS**	**Draft Concl.**	**Final Concl.**
Will all monitored data required for verification and issuance be kept for two years after the end of the crediting period or the last issuance of CERs, for this project activity, whichever occurs later?	**/1/** /10/	DR	Yes.		OK
Monitoring of Project Emissions *It is established whether the monitoring plan provides for reliable and complete project emission data over time.*					
Does the monitoring plan provide for the collection and archiving of all relevant data necessary for estimation or measuring the greenhouse gas emissions within the project boundary during the crediting period?	**/1/** /10/	DR	There are no emissions from the project activity.		OK
Monitoring of Baseline Emissions *It is established whether the monitoring plan provides for reliable and complete baseline emission data over time.*					
Does the monitoring plan provide for the collection and archiving of all relevant data necessary for determining baseline emissions during the crediting period?	**/1/** /10/	DR	The project uses the *ex-ante* determination of emission factor for grid electricity. Only electricity supplied to the grid will be monitored and double checked with the invoice of electricity sold to the grid.		OK
Are the choices of baseline GHG indicators reasonable and conservative?	**/1/** /10/	DR	The choice of baseline indicators is in line with ACM0002.		OK
Is the measurement *method* clearly stated for each baseline indicator to be monitored and also deemed appropriate?	**/1/** /10/	DR	Yes. The electricity meters that used for monitoring are located at the transformer		OK

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			station. The meters will be properly configured and checked annually according to the requirement from Technical administrative code of electric energy metering (DL/T448 －2000). The responsible entity that undertakes the measurement has been specified.		
Is the measurement *equipment* described and deemed appropriate?	**/1/**	DR	Yes. The measurement will be taken through electricity meters.		OK
Is the measurement *accuracy* addressed and deemed appropriate? Are procedures in place on how to deal with erroneous measurements?	**/1/**	DR	Yes. The metering equipment are calibrated and checked annually to ensure accuracy. Procedures to deal with erroneous measurements have been put in place.		OK
Is the measurement *interval* for baseline data identified and deemed appropriate?	**/1/**	DR	The electricity supplied to the grid will be measured continuously and recorded monthly.		OK
Is the *registration, monitoring, measurement* and *reporting* procedure defined?	**/1/**	DR	Yes.		OK
Are procedures identified for *maintenance* of monitoring equipment and installations? Are the calibration intervals being observed?	**/1/**	DR	Yes.		OK
Are procedures identified for day-to-day records handling (including what records to keep, storage area of records and how	**/1/**	DR	Procedures for day-to-day records handling have been identified.		OK

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CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	Ref.	MoV*	COMMENTS	Draft Concl.	Final Concl.
to process performance documentation)					
Monitoring of Leakage *It is assessed whether the monitoring plan provides for reliable and complete leakage data over time.*					
Does the monitoring plan provide for the collection and archiving of all relevant data necessary for determining leakage?	**/1/** /10/	DR	Project participants do not need to consider leakage in applying this methodology.		OK
Monitoring of Sustainable Development Indicators/ Environmental Impacts *It is assessed whether choices of indicators are reasonable and complete to monitor sustainable performance over time.*					
Is the monitoring of sustainable development indicators/ environmental impacts warranted by legislation in the host country?	**/1/**	DR	DNA of China does not require collection and archiving of data related to environmental, social and economic impacts. The environmental impacts will be monitored by local environmental authority.		OK
Does the monitoring plan provide for the collection and archiving of relevant data concerning environmental, social and economic impacts?	**/1/** /5/	DR	The indicators of environmental impacts will be stipulated by local environmental authority.		OK
Are the sustainable development indicators in line with stated national priorities in the Host Country?	**/1/** /5/	DR	Yes. This will be on local authority decision.		OK
Project Management Planning					

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CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	**Ref.**	**MoV***	**COMMENTS**	**Draft Concl.**	**Final Concl.**
It is checked that project implementation is properly prepared for and that critical arrangements are addressed.					
Is the authority and responsibility of overall project management clearly described?	**/1/**	DR I	Yes. The authority and responsibility of overall project management is clearly described.		OK
Are procedures identified for training of monitoring personnel?	**/1/**	DR I	Procedures for monitoring personnel training have been identified.		OK
Are procedures identified for emergency preparedness for cases where emergencies can cause unintended emissions?	**/1/**	DR I	No emergency situation which can cause unintended emissions is expected from the project.		OK
Are procedures identified for review of reported results/data?	**/1/**	DR I	Procedures for internal audit on reading and reporting have been identified.		OK
Are procedures identified for corrective actions in order to provide for more accurate future monitoring and reporting?	**/1/**	DR I	Procedures for corrective actions in order to provide for more accurate future monitoring and reporting have not been established.		OK
C. Duration of the Project/ Crediting Period *It is assessed whether the temporary boundaries of the project are clearly defined.*					
Are the project's starting date and operational lifetime clearly defined and evidenced?	**/1/**	DR I	The project's start date is defined as 10 April 2006, the day on which the construction permit was obtained. The estimated lifetime of the project is 20 years.		OK

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CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	**Ref.**	**MoV***	**COMMENTS**	**Draft Concl.**	**Final Concl.**
Is the start of the crediting period clearly defined and reasonable?	**/1/**	DR I	A renewable crediting period (7 years) is selected starting from 1 March 2008.		OK
D. Environmental Impacts *Documentation on the analysis of the environmental impacts will be assessed, and if deemed significant, an EIA should be provided to the validator.*					
Has an analysis of the environmental impacts of the project activity been sufficiently described?	**/1/** /5/	DR I	Yes. The environmental impacts during construction duration and operation are elaborated in the PDD, mainly about impacts of waste water, noise and solid waste on environment.		OK
Are there any Host Party requirements for an Environmental Impact Assessment (EIA), and if yes, is an EIA approved?	**/1/** /5/	DR I	Yes. The project has been approved by the Environmental Protection Bureau of Inner Mongolia Autonomous Region.		OK
Will the project create any adverse environmental effects?	**/1/** /5/	DR I	There are no significant adverse environmental effects for the project according to the EIA.		OK
Are transboundary environmental impacts considered in the analysis?	**/1/** /5/	DR I	There are no transboundary environmental impacts foreseen for the project.		OK
Have identified environmental impacts been addressed in the project design?	**/1/** /5/	DR I	Yes.		OK
Does the project comply with environmental legislation in the host country?	**/1/**	DR	Yes.		OK

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CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	**Ref.**	**MoV***	**COMMENTS**	**Draft Concl.**	**Final Concl.**
	/5/	I			
E. Stakeholder Comments *The validator should ensure that stakeholder comments have been invited with appropriate media and that due account has been taken of any comments received.*					
Have relevant stakeholders been consulted?	**/1/** /7/	DR	Yes. Besides the stakeholder consultation process required by Chinese EIA regulations, an additional stakeholder consultation process have been performed through inviting different stakeholders comment the project activity.		OK
Have appropriate media been used to invite comments by local stakeholders?	**/1/** /7/	DR I	On 26 July 2006, the project owner held a stakeholders' meeting under the support of the local government. Representatives from the local Government, local Environmental Protection Bureau, local Development and Reform Bureau, local Agricultural Power Bureau and Yangshugoumen Village in Wengniute County attended the meeting.		OK
If a stakeholder consultation process is required by regulations/laws in the host country, has the stakeholder consultation process been carried out in accordance with such regulations/laws?	**/1/** /7/	DR I	Yes. The stakeholder consultation process is in accordance with Chinese EIA regulations.		OK
Is a summary of the stakeholder comments received provided?	**/1/** /7/	DR	Yes. A summary of the stakeholder comments is integrated the PDD.		OK

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CHECKLIST QUESTION * MoV = Means of Verification, DR= Document Review, I= Interview	**Ref.**	**MoV***	**COMMENTS**	**Draft Concl.**	**Final Concl.**
Has due account been taken of any stakeholder comments received?	**/1/** /7/	DR I	No negative comment to the project was received during the consultation.		OK

Table 3 Resolution of Corrective Action and Clarification Requests

Draft report clarifications and corrective action requests by validation team	**Ref. to checklist question in table 2**	**Summary of project owner response**	**Validation team conclusion**
CAR 1 The letters of approval from the DNA of China and the United Kingdom have not been obtained.	A	LoAs from both parties have been received.	OK. DNA of China issued the LoA on 13 December 2006. DNA of the United Kingdom issued the LoA on 13 April 2007. The CAR is closed.
CL 1 Evidence that the incentive from the CDM was seriously considered in the decision to proceed with the project activity needs to be provided.	B	A loan permission letter by China Development Bank issued on 9 March 2006 considered CDM as one of the factors to loan. The CDM benefits were already considered at the time of developing the feseablitity study report. This is confirmed n the approval letter of the feasibility study report of the project by the Development and Reform Commission of Inner Mongolia Autonomous Region on 18 November 2005.	OK. It has been demonstrated that the project owner seriously considered CDM in the decision to develop the project, which has been confirmed in the approval letter of the feasibility study report of the project by the Development and Reform Commission of Inner Mongolia Autonomous Region on 18 November 2005 /4/. The loan permission letter has been verified by DNV. It indicates that the incentive from CDM was seriously considered in the decision to develop the project.

APPENDIX B

CERTIFICATES OF COMPETENCE

CERTIFICATE OF COMPETENCE

Michael Lehmann

Qualification in accordance with DNV's Qualification scheme for CDM/JI (ICP-9-8-i1-CDMJI-i1

GHG Auditor:	Yes		
CDM Validator:	Yes	***JI Validator:***	--
CDM Verifier:	Yes	***JI Verifier:***	--
Industry Sector Expert for Sectoral Scope(s):	Sectoral scope 1, 2, 3		
Technical Reviewer for (group of) methodologies:			
ACM0001, AM0002, AM0003, AM0010, AM0011, AM0012, AMS-III.G	Yes	*AM0027*	Yes
ACM002, AMS-I.A-D, AM0019, AM0026, AM0029, AM0045	Yes	*AM0030*	Yes
ACM003, ACM0005, AM0033, AM0040	Yes	*AM0031*	Yes
ACM0004, ACM0012	Yes	*AM0032*	Yes
ACM0006, AM0007, AM0015, AM0036, AM0042	Yes	*AM0035*	Yes
ACM0007	Yes	*AM0038*	Yes
ACM0008	Yes	*AM0041*	Yes
ACM0009, AM0008, AMS-III.B	Yes	*AM0034*	Yes
AM0006, AM0016, AMS-III.D, ACM0010	Yes	*AM0043*	
AM0009, AM0037	Yes	*AM0046*	
AM0013, AM0022, AM0025, AM0039, AMS-III.H, AMS-III.I	Yes	*AM0047*	
AM0014	Yes	*AMS-II.A-F, AM0044*	Yes
AM0017	Yes	*AMS-III.A*	Yes
AM0018	Yes	*AMS-III.E, AMS-III.F*	Yes
AM0020	Yes		
AM0021, AM0028, AM0034, AM0051	Yes		
AM0023	Yes		
AM0024	Yes		

Høvik, 5 February 2007

Einar Telnes
Director, International Climate Change Services

Michael Lehmann
Technical Director

CERTIFICATE OF COMPETENCE

Mindy (Ming) Yue

Qualification in accordance with DNV's Qualification scheme for CDM/JI (ICP-9-8-i1-CDMJI-i1

GHG Auditor:	Yes		
CDM Validator:	Yes	***JI Validator:***	--
CDM Verifier:	--	***JI Verifier:***	--
Industry Sector Expert for Sectoral Scope(s):	--		

Høvik, 5 January 2007

Einar Telnes
Director, International Climate Change Services

Michael Lehmann
Technical Director

CERTIFICATE OF COMPETENCE

Weidong Yang

Qualification in accordance with DNV's Qualification scheme for CDM/JI (ICP-9-8-i1-CDMJI-i1

GHG Auditor:	Yes		
CDM Validator:	--	***JI Validator:***	--
CDM Verifier:	--	***JI Verifier:***	--
Industry Sector Expert for Sectoral Scope(s):	--		

Høvik, 6 November 2006

Einar Telnes
Director, International Climate Change Services

Michael Lehmann
Technical Director

CERTIFICATE OF COMPETENCE

Tim Kuo

Qualification in accordance with DNV's Qualification scheme for CDM/JI (ICP-9-8-i1-CDMJI-i1

GHG Auditor:	Yes		
CDM Validator:	--	***JI Validator:***	--
CDM Verifier:	Yes	***JI Verifier:***	--
Industry Sector Expert for Sectoral Scope(s):	--		

Høvik, 30 October 2007

Michael Lehmann

Michael Lehmann
Techncal Director, International Climate Change Services

CERTIFICATE OF COMPETENCE

Ole Andreas Flagstad

Qualification in accordance with DNV's Qualification scheme for CDM/JI (ICP-9-8-i1-CDMJI-i1

GHG Auditor:	Yes		
CDM Validator:	Yes	***JI Validator:***	--
CDM Verifier:	--	***JI Verifier:***	--
Industry Sector Expert for Sectoral Scope(s):	--		

Høvik, 30 October 2007

Michael Lehmann

Michael Lehmann
Technical Director, International Climate Change Service

CERTIFICATE OF COMPETENCE

Miguel Rescalvo

Qualification in accordance with DNV's Qualification scheme for CDM/JI (ICP-9-8-i1-CDMJI-i1

GHG Auditor:	Yes		
CDM Validator:	Yes	***JI Validator:***	--
CDM Verifier:	--	***JI Verifier:***	--
Industry Sector Expert for Sectoral Scope(s):	--		
Technical Reviewer for (group of) methodologies:			
ACM0001, AM0002, AM0003, AM0010, AM0011, AM0012, AMS-III.G	Yes		
ACM0002, AMS-I.A-D, AM0019, AM0026, AM0029, AM0045	Yes		
ACM0006, AM0007, AM0015, AM0036, AM0042	Yes		

Høvik, 3 July 2007

Einar Telnes
Director, International Climate Change Services

Michael Lehmann
Technical Director